In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from pandas.core.interchange import column
```

In [2]:
```
#Load Data file
df = pd.read_excel("../data/swiggy_data.xlsx")
print(df)
```

```
           State       City Order Date           Restaurant Name  \
0       Karnataka  Bengaluru 2025-06-29  Anand Sweets & Savouries
1       Karnataka  Bengaluru 2025-04-03     Srinidhi Sagar Deluxe
2       Karnataka  Bengaluru 2025-01-15     Srinidhi Sagar Deluxe
3       Karnataka  Bengaluru 2025-04-17     Srinidhi Sagar Deluxe
4       Karnataka  Bengaluru 2025-03-13     Srinidhi Sagar Deluxe
...           ...        ...        ...                       ...
197425     Sikkim    Gangtok 2025-01-25            Mama's Kitchen
197426     Sikkim    Gangtok 2025-07-02            Mama's Kitchen
197427     Sikkim    Gangtok 2025-03-25            Mama's Kitchen
197428     Sikkim    Gangtok 2025-03-26            Mama's Kitchen
197429     Sikkim    Gangtok 2025-03-27            Mama's Kitchen

                   Location     Category  \
0       Rajarajeshwari Nagar        Snack
1                   Kengeri  Recommended
2                   Kengeri  Recommended
3                   Kengeri  Recommended
4                   Kengeri  Recommended
...                     ...          ...
197425              Gangtok        Momos
197426              Gangtok        Momos
197427              Gangtok        Momos
197428              Gangtok        Momos
197429              Gangtok        Momos

                                           Dish Name  Price (INR)  \
0                                 Butter Murukku-200gm        133.9
1                                           Badam Milk         52.0
2                                       Chow Chow Bath        117.0
3                                          Kesari Bath         65.0
4                                           Mix Raitha        130.0
...                                                ...          ...
197425  Soya cheese chilli momo                    ...        112.0
197426  Kurkure momo fried                         ...        140.0
197427                               Chilli cheese momo        126.0
197428                                  Veg Momos (8 Pc)         85.0
197429                                         Soya Momo        100.0

        Rating  Rating Count
0          4.0             0
1          4.5            25
2          4.7            48
3          4.6            65
4          4.0             0
...        ...           ...
197425     4.4             0
197426     4.4             0
197427     4.4             0
197428     4.4             0
197429     4.4             0

[197430 rows x 10 columns]
```

In [3]:
```
df.head()
```

Out[3]:

| | State | City | Order Date | Restaurant Name | Location | Category | Dish Name | Price (INR) | Rating | Rating Count |
|---|---|---|---|---|---|---|---|---|---|---|
| **0** | Karnataka | Bengaluru | 2025-06-29 | Anand Sweets & Savouries | Rajarajeshwari Nagar | Snack | Butter Murukku-200gm | 133.9 | 4.0 | 0 |
| **1** | Karnataka | Bengaluru | 2025-04-03 | Srinidhi Sagar Deluxe | Kengeri | Recommended | Badam Milk | 52.0 | 4.5 | 25 |
| **2** | Karnataka | Bengaluru | 2025-01-15 | Srinidhi Sagar Deluxe | Kengeri | Recommended | Chow Chow Bath | 117.0 | 4.7 | 48 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **3** | Karnataka | Bengaluru | 2025-04-17 | Srinidhi Sagar Deluxe | Kengeri | Recommended | Kesari Bath | 65.0 | 4.6 | 65 |
| **4** | Karnataka | Bengaluru | 2025-03-13 | Srinidhi Sagar Deluxe | Kengeri | Recommended | Mix Raitha | 130.0 | 4.0 | 0 |

In [4]:
```
df.columns
```

Out[4]:
```
Index(['State', 'City', 'Order Date', 'Restaurant Name', 'Location',
       'Category', 'Dish Name', 'Price (INR)', 'Rating', 'Rating Count'],
      dtype='object')
```

In [5]:
```
df.tail()
```

Out[5]:

| | State | City | Order Date | Restaurant Name | Location | Category | Dish Name | Price (INR) | Rating | Rating Count |
|---|---|---|---|---|---|---|---|---|---|---|
| **197425** | Sikkim | Gangtok | 2025-01-25 | Mama's Kitchen | Gangtok | Momos | Soya cheese chilli momo ... | 112.0 | 4.4 | 0 |
| **197426** | Sikkim | Gangtok | 2025-07-02 | Mama's Kitchen | Gangtok | Momos | Kurkure momo fried ... | 140.0 | 4.4 | 0 |
| **197427** | Sikkim | Gangtok | 2025-03-25 | Mama's Kitchen | Gangtok | Momos | Chilli cheese momo | 126.0 | 4.4 | 0 |
| **197428** | Sikkim | Gangtok | 2025-03-26 | Mama's Kitchen | Gangtok | Momos | Veg Momos (8 Pc) | 85.0 | 4.4 | 0 |
| **197429** | Sikkim | Gangtok | 2025-03-27 | Mama's Kitchen | Gangtok | Momos | Soya Momo | 100.0 | 4.4 | 0 |

In [6]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 197430 entries, 0 to 197429
Data columns (total 10 columns):
 #   Column           Non-Null Count   Dtype
---  ------           --------------   -----
 0   State            197430 non-null  object
 1   City             197430 non-null  object
 2   Order Date       197430 non-null  datetime64[ns]
 3   Restaurant Name  197430 non-null  object
 4   Location         197430 non-null  object
 5   Category         197430 non-null  object
 6   Dish Name        197430 non-null  object
 7   Price (INR)      197430 non-null  float64
 8   Rating           197430 non-null  float64
 9   Rating Count     197430 non-null  int64
dtypes: datetime64[ns](1), float64(2), int64(1), object(6)
memory usage: 15.1+ MB
```

In [7]:
```
df.describe()
```

Out[7]:

| | Order Date | Price (INR) | Rating | Rating Count |
|---|---|---|---|---|
| **count** | 197430 | 197430.000000 | 197430.000000 | 197430.000000 |
| **mean** | 2025-05-01 19:41:20.996808960 | 268.512920 | 4.341582 | 28.321805 |
| **min** | 2025-01-01 00:00:00 | 0.950000 | 1.500000 | 0.000000 |
| **25%** | 2025-03-01 00:00:00 | 139.000000 | 4.300000 | 0.000000 |
| **50%** | 2025-05-02 00:00:00 | 229.000000 | 4.400000 | 2.000000 |
| **75%** | 2025-07-01 00:00:00 | 329.000000 | 4.500000 | 15.000000 |
| **max** | 2025-08-31 00:00:00 | 8000.000000 | 5.000000 | 999.000000 |
| **std** | NaN | 219.338363 | 0.422585 | 87.542593 |

In [8]:
```
#check any null values
df.isnull().sum()
```

Out[8]:
```
State              0
City               0
```

```
Order Date         0
Restaurant Name    0
Location           0
Category           0
Dish Name          0
Price (INR)        0
Rating             0
Rating Count       0
dtype: int64
```

In [9]:

```
#numerical and categorical columns
df['Rating'] = df['Rating'].fillna(df['Rating'].mean())
df['City'] = df['City'].fillna(df['City'].fillna("Unknown"))
```

In [10]:

```
#Checking and dropping duplicate values
df = df.drop_duplicates()
df['City'].duplicated().sum()
df['Dish Name'].duplicated().sum()
```

Out[10]:

```
np.int64(138339)
```

In [11]:

```
#Checking & fixing  Data types
df['Order Date'] = pd.to_datetime(df['Order Date'])
df['Price (INR)'] = df['Price (INR)'].astype(float)
```

In [12]:

```
#Renaming column name
df = df.rename(columns={'Price (INR)': 'Price '})
df.columns
```

Out[12]:

```
Index(['State', 'City', 'Order Date', 'Restaurant Name', 'Location',
       'Category', 'Dish Name', 'Price ', 'Rating', 'Rating Count'],
      dtype='object')
```

In [13]:

```
#Order Date is not enough so to perform monthly yearly wise data
df['Month'] = df['Order Date'].dt.month
df['Day'] = df['Order Date'].dt.day_name()
df['Quarter'] = df['Order Date'].dt.quarter
df['Week'] = df['Order Date'].dt.isocalendar().week
df['Year'] = df['Order Date'].dt.year
```

KPI

In [14]:

```
Total_Sales = df['Price '].sum
Average_Rating = df['Rating'].mean
Average_Order_Value = df['Price '].mean
Total_Orders = df.shape[0]
Rating_Count = df['Rating Count'].count()
print(f"Total_Sales : {Total_Sales}")
print(f"Average_Rating : {Average_Rating}")
print(f"Average_Order_Value : {Average_Order_Value}")
print(f"Total_Orders : {Total_Orders}")
print(f"Rating_Count : {Rating_Count}")
```

```
Total_Sales : <bound method Series.sum of 0         133.9
1          52.0
2         117.0
3          65.0
4         130.0
          ...
197425    112.0
197426    140.0
197427    126.0
197428     85.0
197429    100.0
Name: Price , Length: 197403, dtype: float64>
Average_Rating : <bound method Series.mean of 0         4.0
1         4.5
2         4.7
3         4.6
4         4.0
         ...
197425    4.4
197426    4.4
197427    4.4
197428    4.4
197429    4.4
Name: Rating, Length: 197403, dtype: float64>
```

```
Average_Order_Value : <bound method Series.mean of 0         133.9
1          52.0
2         117.0
3          65.0
4         130.0
          ...
197425    112.0
197426    140.0
197427    126.0
197428     85.0
197429    100.0
Name: Price , Length: 197403, dtype: float64>
Total_Orders : 197403
Rating_Count : 197403
```

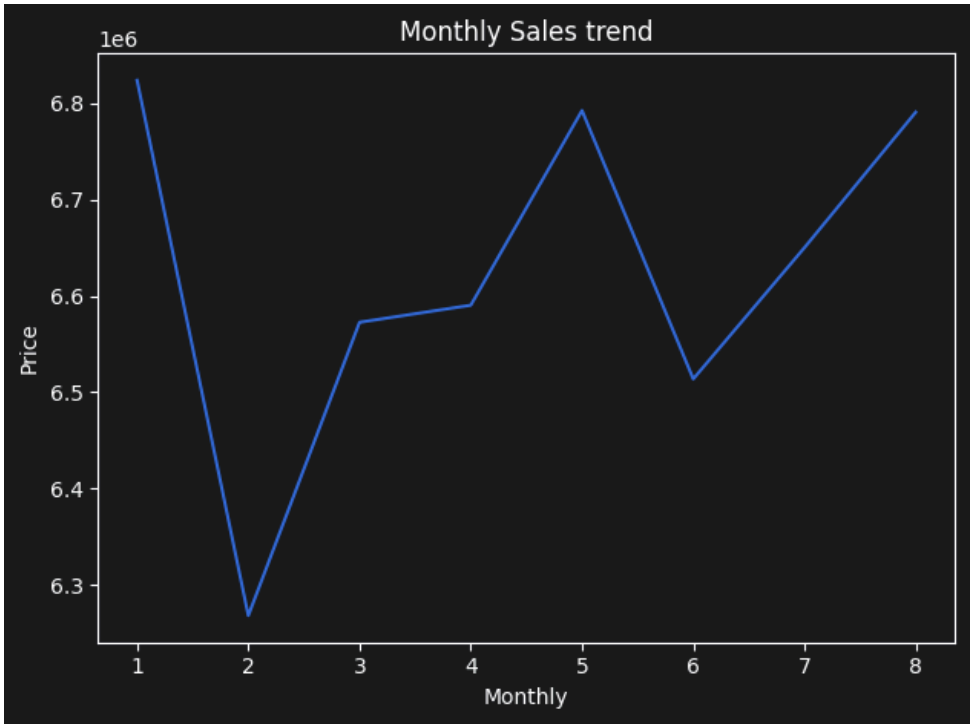
Monthly sales

In [15]:
```
df.columns
```

Out[15]:
```
Index(['State', 'City', 'Order Date', 'Restaurant Name', 'Location',
       'Category', 'Dish Name', 'Price ', 'Rating', 'Rating Count', 'Month',
       'Day', 'Quarter', 'Week', 'Year'],
      dtype='object')
```

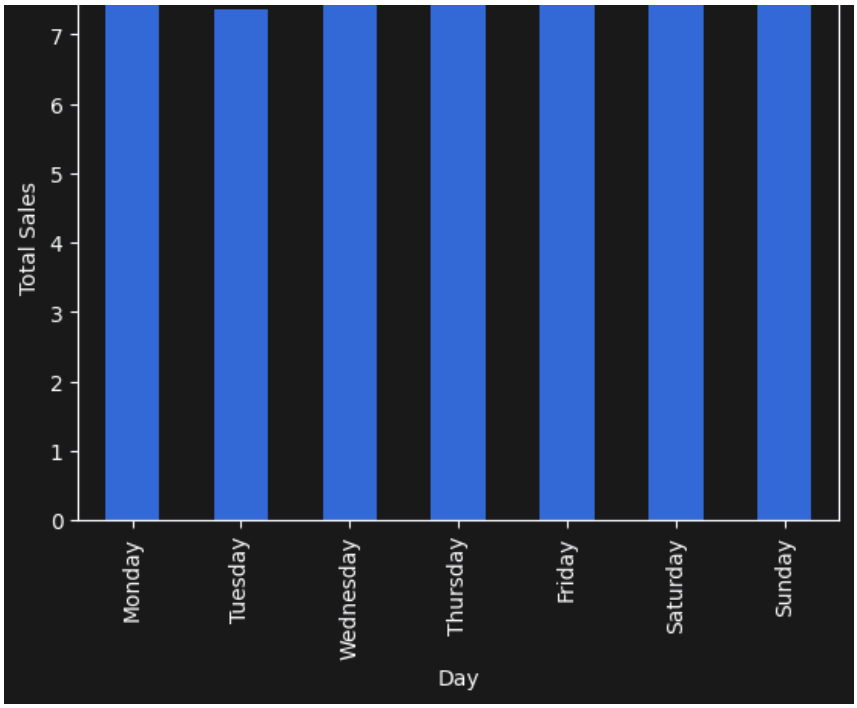
In [16]:
```
monthly_sales = df.groupby('Month')['Price '].sum()
monthly_sales.plot(kind ='line')
plt.title('Monthly Sales trend')

plt.xlabel('Monthly')
plt.ylabel('Price')
plt.tight_layout()
plt.show()
```




Daily Sales Trend

In [17]:
```
daily_sales = df.groupby('Day')['Price '].sum()
order = ['Monday','Tuesday','Wednesday','Thursday','Friday','Saturday','Sunday']
daily_sales = daily_sales.reindex(order)
daily_sales.plot(kind='bar')
plt.title('Daily Sales trend')
plt.xlabel('Day')
plt.ylabel('Total Sales')
plt.xticks(rotation=90)
plt.show()
plt.tight_layout()
plt.show()
```



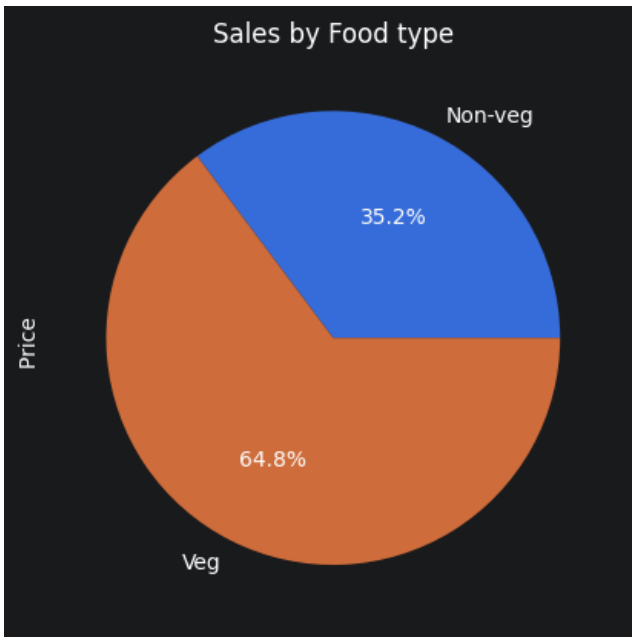

```
<Figure size 640x480 with 0 Axes>
```

Total Sales by Food Types

In [18]:

```
non_veg_keywords = ['chicken','mutton','egg','fish','prawn','biryani','kebab']
df['food type'] = df['Dish Name'].str.lower().apply(lambda x: 'Non-veg'if any(word in x for word in

#Calculating total food sales
food_sales = df.groupby('food type')['Price '].sum()
food_sales.plot(kind='pie',autopct='%1.1f%%')
plt.title('Sales by Food type')
plt.show()
```




In [19]:

```
df.columns
```

Out[19]:

```
Index(['State', 'City', 'Order Date', 'Restaurant Name', 'Location',
       'Category', 'Dish Name', 'Price ', 'Rating', 'Rating Count', 'Month',
       'Day', 'Quarter', 'Week', 'Year', 'food type'],
      dtype='object')
```

Revenue by cuisine

In [20]:

```
cuisine_sales = df.groupby('Category')['Price '].sum().sort_values(ascending=False)
print(cuisine_sales)
```

```
Category
Recommended                     7188272.53
Main Course                      760045.00
Burger Combos ( 3 Pc Meals )     507773.55
McSaver Combos (2 Pc Meals)      431336.65
Desserts                         416172.28
                                   ...
Veg Bread                           25.00
Water                               20.00
Add-ons                             16.95
Cool Drinks                         14.00
Bevarage                            10.00
Name: Price , Length: 4972, dtype: float64
```

In [ ]:

```
cuisine_sales.plot(kind='bar')
plt.title('Revenue by Cuisine')
plt.xlabel('Cuisine')
plt.ylabel('Revenue')
plt.xticks(rotation=0)
plt.show()
plt.tight_layout()
```

```
/Users/akhilmadanu/PycharmProjects/.venv/lib/python3.9/site-packages/IPython/core/pylabtools.py:152:
UserWarning: Glyph 157 (\x9d) missing from font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
```

In [ ]:

```
state_sales = df.groupby('State')['Price '].sum().sort_values(ascending=False)
state_sales.plot(kind='bar',color='skyblue')
plt.title('Revenue by State')
plt.xlabel('State')
plt.ylabel('Revenue')
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```

In [ ]:

```
#Quarterly wise performance summary
df['Quarter'] = df['Order Date'].dt.to_period('Q')
#Group by Quarter
Quarter_sales = df.groupby('Quarter').agg( Total_Sales=('Price ', 'sum'),
                                           Avg_Rating=('Rating', 'mean'),
                                             Total_Orders=('Order Date', 'count'))
Quarter_sales.plot(kind='bar')
plt.title('Quarter sales Trend')
plt.xlabel('Quarter')
plt.ylabel('Values')
plt.xticks(rotation=0)
plt.show()
plt.tight_layout()
```